

# Cao Thang Nguyen

✉thangckt@gmail.com | 🌐thangckt.github.io | 🎓scholar.google

## TECHNICAL SKILLS

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### Computational Modeling & Simulation

- Molecular Dynamics (MD) and Molecular Statics (MS) simulations.
- Density Functional Theory (DFT) calculations for materials modeling
- Free energy calculations using Enhanced Sampling methods (Steered MD, MetaDynamics, Mean-Force Dynamics) and using Thermodynamic Integration (TI).
- Computational modeling of crystalline and polymeric molecular systems.
- Application of Machine Learning in computational materials science.
- Strong background in Finite Element Method (FEM), Solid Mechanics, Statistical Mechanics, Thermodynamics.

### High-Performance Computing & Automation

- Building and maintaining high-performance computing (HPC) clusters.
- Compiling, configuring, and running large-scale simulation packages (e.g., LAMMPS, PLUMED, GPAW, etc.) on HPC clusters.
- Develop computational pre- and post-processing pipelines.
- Develop automated simulation workflows across heterogeneous HPC environments and job schedulers.

### Programming & Software

- Programming Languages: Python, Matlab, C++ (Basic), Bash script.
- Computational codes/packages: LAMMPS, PLUMED, GPAW, ASE, DFTD3, OVITO, Abaqus FEA.
- Scientific libraries: NumPy, SciPy, Pandas, Polars, Matplotlib, Pytorch, etc.
- Reproducibility tools: Git/GitHub/GitLab, GitHub Actions, Jupyter Notebook, Conda, Pip, LaTeX
- OS: CentOS, Rocky Linux, Fedora, Ubuntu, Windows, WSL

## EXPERIENCES

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**Postdoctoral Researcher** – UNIST, South Korea Feb 2023 – Present

- Developing Machine Learning Interatomic Potentials (MLIPs) for TMDCs materials using GNN-based architectures.
- Developing automated framework for active learning of MLIPs and material properties calculation.

**Graduate Researcher** – UNIST, South Korea Mar 2017 – Feb 2023

- Calculated free energy of solid surface and solid-liquid interfaces using enhanced sampling methods.
- Investigated melting behavior of metallic structures at the nanoscale.

**Graduate Researcher** – University of Ulsan, South Korea Sep 2014 – Aug 2016

- Investigated the elastic instabilities of metals and metallic porous structures under various loading conditions using Molecular Dynamics Simulations.

**R&D Staff** – Industrial Zones, Vietnam Jan 2010 – Sep 2014

- Developing and stabilizing manufacturing processes.
- Technical support for customers.

## SELECTED PROJECTS

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**ALFF**: Automated Frameworks for Active-Learning Machine Learning Interatomic Potentials and Material Properties Calculation.

- This package automates the generation of training data and the refinement of GNN-based architectures (such as MACE, SevenNet, etc.) through active learning cycles.
- Automates the computation of material properties, including elastic constants, phonon dispersions, PES, etc.
- A distinctive capability of ALFF is its ability to distribute workloads across multiple remote clusters, enabling the combination of different computing resources – such as Web Services clouds, National Supercomputing centers, local campus clusters – and having them collaborate seamlessly within a single workflow.

Contributed to many open-source packages, including [DPGEN](#), [dpdata](#), [DPDispatcher](#), [ASE](#), [mBuild](#), etc.

## PUBLICATIONS

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10+ publications in peer-reviewed journals, please refer to [Scholar profile](#) for details.

## EDUCATION

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**Ulsan National Institute of Science and Technology (UNIST)** Ulsan, South Korea  
Ph.D. in Mechanical Engineering Mar 2017 – Feb 2023

**University of Ulsan** Ulsan, South Korea  
M.S. in Mechanical Engineering Sep 2014 – Aug 2016

**Ho Chi Minh City University of Technology and Engineering** Ho Chi Minh City, Vietnam  
B.S. in Mechanical Engineering Sep 2005 – May 2010

## NOTES

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- I am a lawful permanent resident, authorized to work in the United States without need for sponsorship.